BrickPlan 1.1: Certified Assembly-Order Diagnostics Across Two Source Families

Executable benchmark release

October 2, 2026

Abstract

A correctly placed part can obstruct a future insertion and force removal. BrickPlan diagnoses this planning failure with locally goal-independent actions, exact future-feasibility labels, and auditable recovery costs. Version 1.1 extends the frozen 1.0 bank with an original airframe subassembly from a second source kit. The bank contains 3 structures, 12 episodes, and 39 diagnostic responses, covering 342,704 reachable states and 2,843,080 directed transitions. The new structure requires 21 attachments from its fixed base and has an exact mandatory chain of 5 parts across all detach-free completions. Seven dimensions measure dead-end choices, supplied-plan conflict prediction, plan survival, regret, recovery, replanning efficiency, and counterfactual consistency. Every injected repair requires three removals. A shared HTTP executor supports human practice, model clients, and original-mesh 3D replay. Results below are software validation controls; no model or human performance has been measured.

1 Problem and contribution

Assembly planning must maintain object identities, support, access paths, and future requirements across many individually plausible actions. Closing a region early may execute successfully while making every remaining attachment-only completion impossible. End-to-end success cannot distinguish this event from an immediately invalid action or an inaccurate pose.

LEGO-Puzzles motivates multistep planning evaluation [1]; PhyBlock studies physical planning [2]; BrickComposer studies fine-grained selection and placement [3]. Their reported metrics measure different outcomes and are not interchangeable episode success rates. Interactive assembly and reconstruction also precede this benchmark, including LTRON [4] and InstructioNet [5]. The contribution here is an executable measurement contract connecting decisions, feasibility loss, diagnosis, and recovery to exact finite-domain certificates. It is not a claim that interaction or disassembly is new.

The extension addresses two concrete limitations of 1.0: its two structures shared one source kit, and both mandatory chains contained four parts. The airframe adds a separate kit and a longer certified order. Two provenance groups still do not establish broad generalization, and a deeper mandatory assembly chain does not establish deeper conflict-probe causality.

2 Workcell and certified bank

2.1 Action and observation contract

Each task supplies a connected, clamped fixed base, an inventory, calibrated sites, and a visible exact target. Original source geometry and placements are preserved. Attach and detach move one part along its declared straight path. Every accepted state remains connected to the fixed base through audited stud connections, abstracted as rigid holds. External magazine transfer, hands, grasping, clutch force, and gravity are outside the certified domain.

Local legality depends only on current occupancy, path obstruction, and support. The executor does not consult the future goal or cost oracle. Thus a legal closure that causes later rework actually executes. The goal requires exact occupancy, including removal of extra parts. All sites remain selectable; there is no next-step feasibility mask. Structural attempts cost one action, including rejected attempts; inspection and diagnosis have separate quotas.

Public part and site IDs are stable within an episode and randomized by seed. Equivalent parts of the same type and color are exchangeable. Observations provide source geometry in calibrated coordinates but exclude instruction indices, blocker tables, oracle costs, viability, and reference actions. The browser adds camera control and an exploded reference for practice and review. Actual assembly states retain their source poses.

2.2 Geometry and provenance

Display uses original recursively resolved LDraw triangles. Collision proxies are closed meshes inset by 0.25 LDU per surface, with a 0.002 LDU guard. All final-pose pairs must be clear. Each moving part is checked against every other included part across its full declared translation. Separating swept bounds certify easy clear pairs; otherwise midpoint separation d certifies an interval $[l, h]$

when $d > (h - l)/2 + 0.002$, by the 1-Lipschitz property of surface distance under unit translation. Triangle intersections provide blocked witnesses. Unresolved coverage is UNKNOWN and excludes the candidate.

The inherited 3022 collision-proxy repair fills one source-verified missing triangular stud-cap face without moving vertices. Original render meshes remain unchanged. These are numerical certificates for the declared proxy/path domain, not guarantees for unrestricted six-degree-of-freedom manipulation.

The airframe selects source instances 0–34 excluding 11–14 and 21–27; instances 1, 2, and 5 form its fixed base. The propeller, float/clip assemblies, and main wing plates are outside this fuselage, tail and upper-plate subassembly. The 54383/54384 wing proxies contain extensive nonmanifold and boundary edges; both wings and their dependent top plate are excluded without proxy repair. Every included endpoint and motion pair is requalified; earlier exploratory sampled paths are not accepted as ground truth. The exact new mandatory chain is 3 $\rightarrow$ 8 $\rightarrow$ 16 $\rightarrow$ 30 $\rightarrow$ 32 in private source-index notation. These source indices never enter policy observations.

2.3 Exact finite-domain labels

Let s encode occupied mutable sites and g the exact goal. Exhaustive search enumerates the reachable component; goal-rooted breadth-first search computes $C^*(s, g)$, the minimum successful unit-action count. Every edge has a verified inverse. For every stored goal, Bellman inequalities, unique zero-cost termination, component equality, and decreasing successors certify the table. Geometry, source, transition, task, and cost hashes identify the release.

If $\Delta = |g| - |s|$, successful one-part actions satisfy attachments minus detachments $= \Delta$. Therefore

$$D^*(s, g) = (C^*(s, g) - \Delta)/2.$$

No-undo feasibility is $V^+(s, g) = [s \subseteq g \wedge C^*(s, g) = \Delta]$; budget feasibility is $V_B(s, g, b) = [C^*(s, g) \leq b]$. A no-undo dead end is recoverable with removal, not absolute impossibility.

Mandatory precedence is the intersection of ordering constraints over *all* detach-free full completions. A source pair $i \prec j$ is mandatory only if no viable detach-free prefix contains j without i. The longest chain in this relation is separate from total task length and from the first-conflict distance of a supplied plan.

2.4 Episodes and paired probes

Each structure supplies full, checkpoint, repair, and matched healthy episodes. Checkpoints are at least three successful actions from the fixed base and offer both safe and costly locally legal continuations. Repair starts are reachable and require exactly three detachments. Healthy controls match installed count and inherit the repair budget. Default budgets are $C^* + \max(4, \lceil C^*/2 \rceil)$.

Per structure, one choice probe evaluates a legal decision at a tight budget; eight lookahead probes cover distances 1, 2, 4, and 8 with executable controls; four counterfactual members form a switching pair and an invariance pair. All starts have reachability witnesses, and all episode completions have optimal action witnesses. Random IDs and repeated runs do not create new independent structures.

3 Seven diagnostic dimensions

Dead-end choice rate. Among eligible legal choices with feasible predecessors, count losses of V^+, and separately losses of V_B after decrementing the budget. Missing, invalid, and valid responses have separate counts; all-assigned avoidance prevents abstention from earning full credit.

Lookahead depth. Predict the first failing action index of a supplied continuation, or null when all actions execute. Report exact accuracy, conflict recall, false alarms, and coverage by distance. The causal depth of these obstruction probes remains *one*, including in 1.1. Longer bins contain intervening legal attachments. An executable control need not preserve eventual goal feasibility. This measures a conflict-prediction curve, not internal search depth.

Plan survival. Freeze the first attempted-action index losing each feasibility predicate; later recovery never erases the loss. For N initially eligible runs, report known survivors A_t and unresolved early stops U_t with bounds $[A_t/N, (A_t + U_t)/N]$. Completed successes are absorbing, while abandoned runs are not assumed to survive indefinitely. Stratify by case and mode.

Regret. Successful action regret is attempted actions minus $C^*(s_0, g)$; detach regret is detach operations minus $D^*(s_0, g)$. Failed or unfinished runs have null regret and explicit consumed costs. In this domain, action regret equals twice detach regret plus failed attempts; these are related cost measures.

Recovery rate. Score the first explicit needs-rework diagnosis before the first successful structural action. Prior failed attempts are separately flagged. Report detection recall, precision, healthy false alarms, recovery completion, and joint detection/completion. Later diagnoses cannot replace the first answer.

Replanning efficiency. Report detaches, distinct removed instances, wasted attempts, regret, and the first

Table 1: Frozen case bank. Roadster and kart share LEGO 31027; airframe is from LEGO 31028. Counts describe separate state graphs, not independent trials.

Case	Source group	Fixed/movable	States	Directed edges	Full actions	Chain parts	Repair detaches
Roadster	LEGO-31027	20/17	14,012	131,824	17	4	3
Kart	LEGO-31027	7/18	44,604	436,824	18	4	3
Airframe	LEGO-31028	3/21	284,088	2,274,432	21	5	3

Table 2: Privileged software controls; not model or human results.

Controller	Complete	Repair
Optimal oracle	12/12	3/3
Fault + recovery	12/12	3/3
Trap + attach only	3/12	0/3

Table 3: Preregistered study framework; empirical slots remain unfilled.

Condition	Status
Model: stepwise structured policy	Not evaluated
Same model: explicit planning	Not evaluated
Human structured protocol	Not evaluated
Matched image-only observations	Not implemented

useful repair reducing D^*. Conditional efficiency accompanies completion rate; self-induced and injected recovery states remain distinct.

Counterfactual consistency. Hold state, inventory, sites, paths, and two offered closures fixed while changing the visible goal. Both closures remain required. Exact conditional costs $Q^*(s, a, g) = 1 + C^*(T(s, a), g)$ determine the best *offered* choice. Switch pairs reverse the unique minimizer; invariance pairs retain it. Score joint correct switching and correct invariance after depermuting candidate positions. A globally better interior insertion may exist outside the two offered actions.

4 Validation and experimental settings

Complete success at the frozen budget is the headline outcome; no weighted composite is defined. The unchanged 1.0 engine and metric implementation are bound to the extended bank through a release-specific entry point. Old files remain frozen dependencies. The new launcher stamps collected metadata and evaluation output with the benchmark version. HTTP requests are revision checked, idempotent, and atomically journaled. Terminal-only exports are replayed authoritatively before scoring.

17 semantic tests verify complete cost tables, mandatory precedence, source preservation, proof tampering, invalid inputs, missing responses, censoring, wrong switching, budget exhaustion, identity, and diagnosis timing. HTTP validation completes 12 episodes and 39 responses, then checks restart, collection, scoring, and private-route rejection. Browser validation executes full and repair episodes for every structure through actual buttons, including a charged incompatible-part attempt in each full episode. It verifies replay, downloads, displayed-probe identity, and desktop/mobile layout.

Before a model study, specify exact model/provider/version, prompts, token and context budgets, reasoning settings, observation access, seeds, retry policy, costs, and selection among repeated runs. Isolate the policy process from private certificate files and post-trial oracle tools. Split by source family; roadster and kart must remain together. Two source groups are insufficient for strong population-level confidence statements.

5 Limitations

The new case strengthens provenance coverage and mandatory dependency length, but all full episodes remain short diagnostic builds in the proposed D1 range. The supplied-plan obstruction probes still have causal depth one. No maximum reasoning horizon, 100-step planning, VLM perception, or human advantage is established. Known poses, fixed straight paths, rigid stud support, and inset meshes restrict physical validity. Alternative insertion directions, simultaneous subassembly motion, free rotation, calibrated clutch forces, hand access, and physical construction remain unverified. The next empirical question is whether these diagnostics explain model failures beyond immediate action errors.

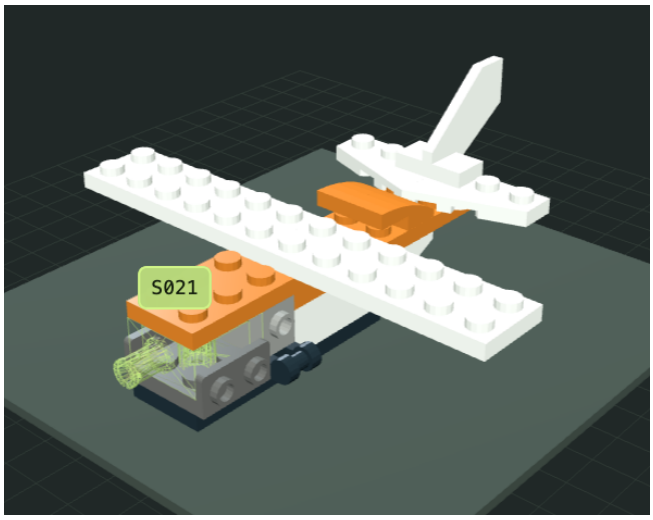
References

[1] LEGO-Puzzles. arXiv:2503.19990, v4. https://arxiv.org/html/2503.19990v4.

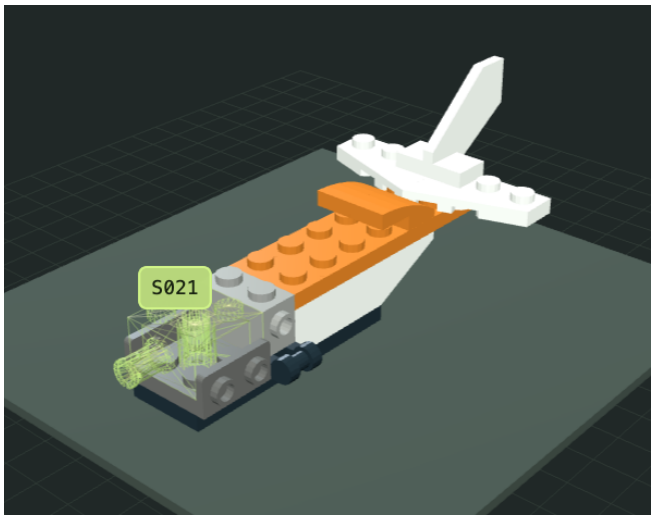
[2] PhyBlock. arXiv:2506.08708, v2. https://arxiv.org/html/2506.08708v2.

[3] Brick-Composer. arXiv:2606.05445, v1. https://arxiv.org/html/2606.05445v1.

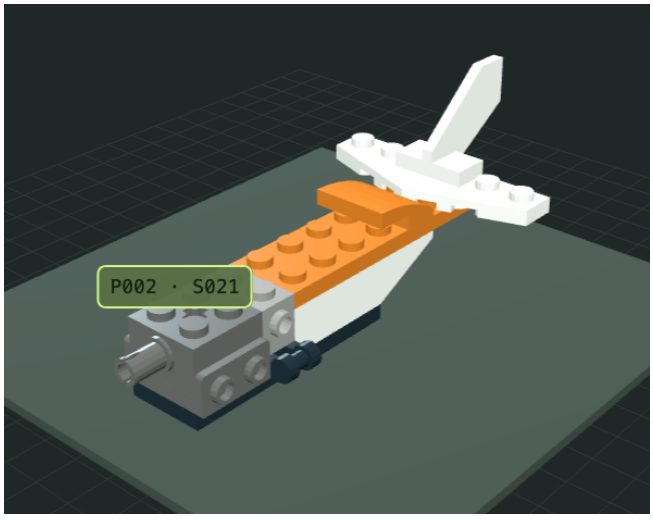
[4] LTRON / Break and Make. arXiv:2207.13738. https://arxiv.org/abs/2207.13738.



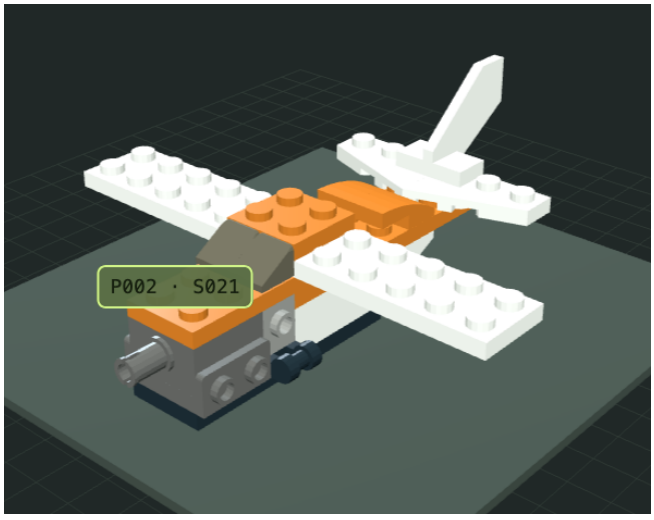
(a) Reachable repair start.



(b) Three removals completed.



(c) Install the interior part.



(d) Restore the target.

Figure 1: Actual airframe repair replay, seed 31. Remove P008, P012, P019, then install P002 at S021 and restore the target. The complete recovery uses 10 successful actions and exactly three detachments, matching the certified optimum. Empty-site wireframes are locators, not installed floating parts. Frames come from authoritative executor states rendered with original LDraw triangles; only screenshot margins are cropped.

[5] InstructioNet. arXiv:2410.01111, v1. https://arxiv.org/html/2410.01111v1.